# Surveyor Supreme, an Interpretable Learning Framework for Survey Engagement and Completion

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# 1 Introduction

This project develops a data-driven internal analytics tool to support survey optimisation and employer-attractiveness analysis using large-scale survey responses. By integrating exploratory data analysis, clustering, and behavioural profiling, the framework transforms raw survey data into actionable and interpretable intelligence for internal survey operations. Correlated and redundant survey items are identified to inform evidence-based survey streamlining, while unsupervised learning is used to uncover meaningful demographic and behavioural segments and analyse differences in employer perception and applicant motivations. Interpretability is prioritised through clear visualisations and structured summaries that enable non-technical users to explore key trends and drivers. Overall, the proposed framework demonstrates how advanced analytics can be operationalised to improve survey effectiveness, enhance respondent experience, and generate insights that inform employer branding and talent-attraction strategies.

# 2 Dataset Overview

The dataset used in this study consists of anonymised survey responses collected as part of the GRADSG survey, which aims to understand students' and graduates' perceptions of employers, career motivations, and information needs. Each record corresponds to a single survey response and contains a mixture of demographic variables, survey metadata, Likert-scale ratings, multiple-choice questions, and free-text inputs. In addition to respondent background information such as institution, year of study, highest qualification, main field of study, nationality, and gender, the dataset includes key outcome variables such as perceived employer attractiveness (rated on a scale of 1 to 10), current perception of the employer, and factors motivating job applications. Several questions adopt a multi-select format, capturing respondents' priorities across dimensions such as career progression, compensation, work-life balance, organisational culture, and the application process. The dataset also contains operational metadata including response start and submission timestamps, completion status, language, device and session information, and geographic indicators. These fields enable the analysis of survey completion behaviour, response patterns, and potential sources of survey friction or dropout.

## 2.1 Challenges

Despite the richness of the GRADSG survey dataset, several analytical and operational challenges complicate the extraction of reliable, actionable insights.

1. **High-Dimensional and Heterogeneous Data**
   The dataset contains a mixture of demographic variables, Likert-scale ratings, multi-select ("pick-3") questions, free-text responses, and survey metadata. This heterogeneity complicates feature encoding and integration, while multi-select questions increase dimensionality and sparsity, making clustering and pattern mining more difficult.

2. **Redundancy and Overlapping Survey Constructs**
   Several survey items capture closely related concepts, particularly across career progression, compensation, culture, and application processes. Strong co-occurrence patterns suggest redundancy and overlapping informational content, creating ambiguity and motivating the need for systematic redundancy detection and survey streamlining.

3. **Partial and Incomplete Responses with Behavioural Heterogeneity**
   A non-trivial proportion of survey sessions are incomplete, introducing missingness that is not completely at random and potentially biasing analysis. At the same time, respondents exhibit wide variation in employer perceptions, motivations, and informational priorities, limiting the usefulness of aggregate analysis and motivating segmentation to uncover distinct respondent personas.

# 3 Methodology

To address the objectives, we propose a multi-stage analytical framework that transforms raw survey responses into actionable insights for survey optimisation and employer-attractiveness analysis. Our methodology integrates data processing, exploratory analysis, pattern mining, unsupervised learning, and automated insight synthesis. The overall pipeline is designed to support four key analytical goals: (i) understanding respondent behaviour, (ii) identifying redundancies and correlations in survey questions, (iii) discovering meaningful respondent segments, and (iv) extracting drivers of employer attractiveness.

## 3.1 Data Processing

To prepare the GRADSG survey data for analysis and modelling, a structured data-processing pipeline was implemented to ensure consistency, usability, and analytical validity.

1. **Data Cleaning and Standardisation**
   Inconsistent labels, trailing whitespace, and formatting differences in categorical variables were standardised. Invalid or ambiguous entries were removed or corrected to ensure uniform category definitions across demographic and survey-response fields. Redundant columns like “Language”, “Tags”, and “SessionID” were removed as they either only had one value in the entire column, only contained null values, or they were not useful in our analysis.

2. **Handling Missing and Incomplete Responses**
   Records were filtered by completion status to support both full-response analysis and dropout behaviour analysis. Missing values were replaced with the column’s mode value.

3. **Feature Engineering and Transformation**
   Derived variables were created to summarise respondent behaviour, such as total selections across question blocks and grouped motivation indicators. Numeric variables such as Attractiveness level were normalised where required to ensure comparability across features.

4. **Text Preprocessing for Free-Response Fields**
   Free-text inputs were lower-cased and cleaned by removing punctuation and stopwords. These fields were prepared for downstream qualitative analysis and theme extraction.

## 3.2 Survey Pattern Mining and Redundancy Detection Methodology

All multi-select ("pick-3") survey questions were transformed into a binary selected / not selected format to ensure a consistent analytical representation across question blocks. Categorical demographic and perceptual variables were standardised and encoded, while numeric variables such as education level were retained in their original scale. Missing values were handled conservatively to preserve partial responses for dropout and behavioural analysis. This preprocessing step produced a unified, analysis-ready dataset suitable for co-occurrence analysis, respondent segmentation, and redundancy detection.

### 3.2.1 Global Multiple Correspondence Analysis (MCA)

A layered multivariate framework is designed to detect structural patterns, cross-question correlations, and redundancies in a multi-question survey where most questions were encoded as binary "selected/not selected" response options. Global MCA was fitted on the combined, filtered dataset to position all categories in a shared latent space, allowing survey owners to visually assess which options cluster together (similar) and which separate (distinct) (Fig 1). For interpretability, emphasis was placed on the "Yes" categories when reviewing the factor structure.

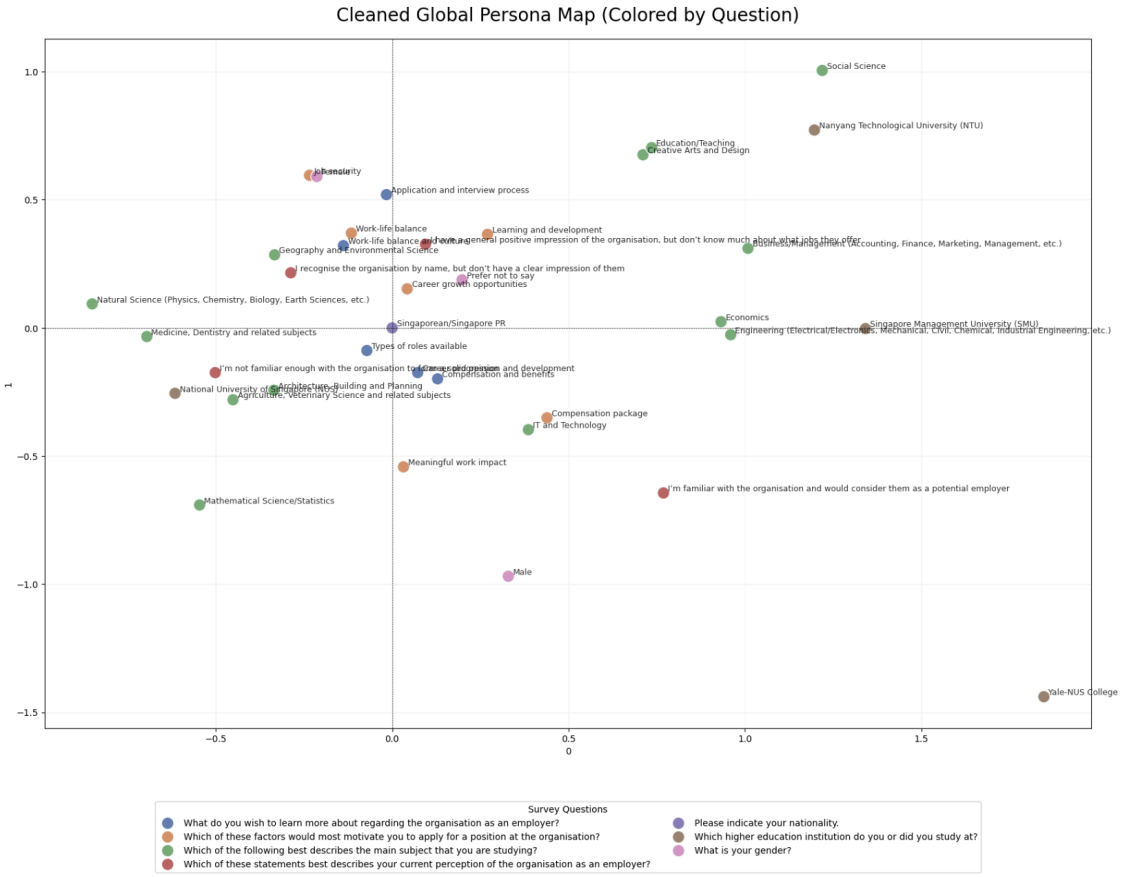


Figure 1

### 3.2.2 Local Correlation Structure within Question Blocks

Local MCA was fitted separately within each question to isolate within-question trade-offs and reveal how options align or oppose one another. To support consistent interpretation, cosine-based similarity scores were computed from local MCA coordinates to quantify option tendencies.

### 3.2.3 Respondent Feature Construction and Segmentation

To translate these latent structures into actionable segmentation, a combined pipeline was implemented. Respondent row coordinates from each local MCA were extracted and concatenated into a single respondent feature set; numeric variables such as education level were appended and all features were standardized to ensure comparability across questions.

Truncated SVD was then applied to reduce the combined feature space into a small set of interpretable components, and K-Means clustering was run on the SVD space to generate respondent personas. Persona labels were derived through lift-based profiling using original categories and numeric indices.

In parallel, a semantic redundancy test was conducted using Sentence Transformer embeddings: option texts were cleaned to remove shared prefixes and compared using cosine similarity to flag near-duplicate constructs.

### 3.2.4 Redundancy Detection and Semantic Similarity Analysis

To support redundancy detection at the textual level, answer option texts were cleaned to remove shared prefixes and boilerplate phrasing. Pairwise cosine similarity between option texts was computed to flag near-duplicate constructs.

These semantic similarity signals were used to reinforce behavioural redundancy detected from association rules and co-occurrence analysis, ensuring that flagged redundancies reflected both behavioural and linguistic overlap.

### 3.2.5 Actionability and Survey Improvement Mapping

Finally, detected redundancies, correlated options, and dominant response patterns were translated into concrete survey improvement recommendations. These included candidate answer-option mergers, question consolidation opportunities, and reordering suggestions to reduce cognitive load and survey fatigue.

This final step ensures that all analytical outputs directly support the operational objective of improving survey design and effectiveness.

## 3.3 Clustering of Geographical Data

### 3.3.1 Overview and Motivation

The dataset contains multiple fields related to respondents' geographical information, namely "**IP Address", "Longitude", "Latitude", "Country", "City", "State/Region", and "Postal"**. While these

columns provide rich location-based context, using them individually in downstream analysis would introduce high dimensionality, redundancy, and noise.

To address this, we aim to reduce these geographical attributes into a single, meaningful cluster label that captures spatial proximity between respondents. This clustered representation allows geographical patterns to be incorporated into later analyses in a compact and interpretable manner.

Given that geographical data is inherently spatial and distance-based, we adopt a **density-based clustering approach**, with a particular focus on Longitude and Latitude.

### 3.3.2 Choice of Clustering Algorithm

**DBSCAN (Density-Based Spatial Clustering of Applications with Noise)** will be chosen as the main clustering algorithm for geographical data. DBSCAN is well-suited for this task for several reasons:

- No need to pre-specify the number of clusters, which is desirable given the unknown spatial distribution of respondents.
- Ability to identify noise points, such as isolated respondents or rare locations.
- Effectiveness on spatial data, especially when paired with a distance metric such as the Haversine distance for latitude–longitude coordinates.

Only Latitude and Longitude are used directly for clustering, as other geographical fields (e.g. country or city) are categorical representations of the same spatial information and would be redundant.

### 3.3.3 Distance Metric and Preprocessing

Latitude and longitude values are first converted from degrees to **radians**, allowing the use of the **Haversine distance**, which measures great-circle distance on the Earth's surface. This ensures that distances between points are geographically meaningful rather than Euclidean approximations.

### 3.3.4 Determining the Optimal Epsilon (ε)

A critical parameter in DBSCAN is **ε (epsilon)**, which defines the maximum distance between two points for them to be considered neighbours. Rather than choosing ε arbitrarily, we adopt a **data-driven approach** based on the *k-distance graph*.

We set:

- minPts = 4, a common heuristic for 2D spatial data
- Hence, $k = \text{minPts} - 1 = 3$

For each data point, we compute the distance to its **3rd nearest neighbour** using the Haversine metric. These distances are then sorted in ascending order to form the k-distance curve.

To automatically identify candidate elbow points (i.e. points of rapid change in distance), we apply a **gradient-based elbow detection algorithm**:

1. Compute the **first and second derivatives** of the sorted k-distance curve.
2. Focus on the upper half of the curve, where elbows are more likely to occur.
3. Identify local maxima in the second derivative that exceed a threshold defined by the mean plus a multiple of the standard deviation.

4. If no strong elbows are detected, fallback epsilon values are chosen based on high-percentile distances (90th and 95th percentiles).

This procedure returns up to **three candidate ε values**, ranked by prominence. An ε value of 0.116806km will be chosen, which will then be divided by 6371km.

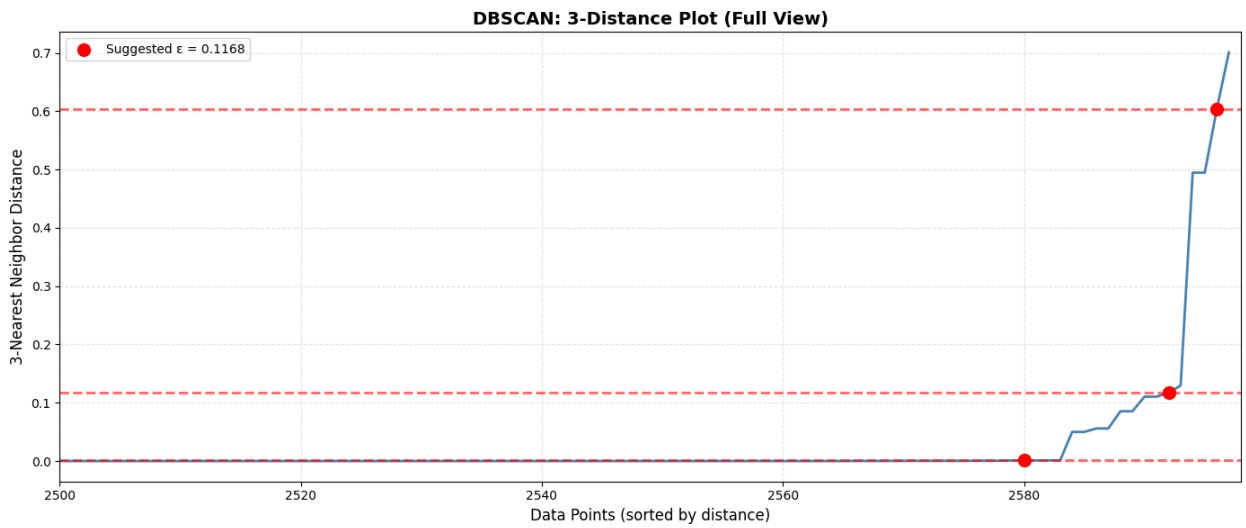


Figure 2

### 3.3.5 Final Clustering and Feature Engineering

After selecting the most appropriate ε value, DBSCAN is run on the full set of latitude–longitude coordinates. The resulting **cluster labels** are then appended to the original dataset as a new feature.

This cluster label serves as a **compressed geographical representation**, effectively replacing multiple raw location columns with a single, informative variable. This enables subsequent analyses to incorporate geographical information without unnecessary complexity.

## 3.4 Partial Response Modelling and Engagement Analysis

### 3.4.1 Problem Formulation and Target Definition

The objective of this analysis is to identify behavioural and perceptual factors associated with partial or incomplete survey responses. Rather than treating non-completion as random attrition, we formulate it as a supervised classification and engagement modelling problem.

Let the survey outcome variable $Y \in \{0,1\}$, where:

$$Y = \begin{cases} 1 & \text{if survey status} = \text{Complete} \\ 0 & \text{if survey status} = \text{Partial or Disqualified} \end{cases}$$

This binary target enables probabilistic modelling of completion likelihood. Two complementary analytical tracks are defined:

1. Perception-based modelling, using respondents' motivation and employer-interest variables to explain completion behaviour.

2. Time-based engagement modelling, using survey duration as a behavioural proxy for respondent engagement.

### 3.4.2 Feature Construction and Preprocessing

Two feature groups were constructed from the raw survey dataset:

(i) Motivation Factors
These include Likert-scale ratings capturing intrinsic and extrinsic drivers such as:

- Career growth opportunities
- Compensation package
- Job security
- Learning and development
- Meaningful work impact
- Work–life balance

(ii) Employer Information Interest Variables
Binary or ordinal variables capturing what respondents wished to learn more about regarding the organisation as an employer, including:

- Types of roles available
- Career progression and development
- Compensation and benefits
- Work–life balance and culture
- Application and interview process

Column names were standardized by removing long question prefixes for consistency. Missing values were imputed with zeros to represent the absence of expressed interest or non-response. All features were standardised using z-score normalisation before modelling to ensure coefficient comparability.

### 3.4.3 Logistic Regression for Completion Probability Modelling

A logistic regression classifier was trained to estimate the probability of survey completion as a function of respondent perceptions and interests.

The model is defined as:

$$P(Y = 1 \mid X) = \sigma(\beta_0 + \beta^T X)$$

where $\sigma(\cdot)$ is the logistic sigmoid function, X is the feature vector, and $\beta$ are the learned coefficients.

To address class imbalance between completed and non-completed surveys, class-balanced weighting was applied. The model was trained using the liblinear solver with a maximum of 2,000 iterations to ensure convergence stability.

Model interpretability was prioritised. Coefficients were exponentiated to obtain odds ratios:

$$\text{Odds Ratio}_i = e^{\beta_i}$$

This transformation enables direct interpretation of the multiplicative effect of each feature on the odds of survey completion.

### 3.4.4 Survey Completion Time Engineering

Survey engagement was operationalised using completion time computed from timestamp differences:

$$\text{Time Taken (seconds)} = \text{Date Submitted} - \text{Time Started}$$

This value was converted into minutes for interpretability. Observations with non-positive durations were removed to eliminate logging artefacts.

To improve interpretability and enable discrete behavioural segmentation, completion time was further bucketed into five ordinal engagement categories:

- < 1 minute
- 1–3 minutes
- 3–10 minutes
- 10–30 minutes
- 30 minutes

These buckets approximate intuitive engagement depth regimes ranging from immediate abandonment to sustained interaction.

### 3.4.5 Non-Parametric Statistical Testing of Engagement Differences

Due to extreme right-skew and heavy-tailed behaviour in completion time distributions, parametric assumptions of normality were violated. Accordingly, non-parametric methods were employed.

A Kruskal–Wallis H-test was used to test the null hypothesis that completion time distributions are identical across the three outcome groups (Complete, Partial, Disqualified).

$$H = \frac{12}{N(N+1)} \sum_{i=1}^{k} n_i (\bar{R}_i - \bar{R})^2$$

where $n_i$ is the group size, $\bar{R}_i$ is the mean rank of group **i**, and k is the number of groups.

Effect size was computed using eta-squared:

$$\eta^2 = \frac{H - k + 1}{N - k}$$

To identify pairwise group differences, Mann–Whitney U-tests were performed between all outcome pairs. Bonferroni correction was applied to control the family-wise error rate.

### 3.4.6 Time-Bucket Composition Analysis

For each engagement bucket, the empirical distribution of survey outcomes was computed:

$$P(\text{Status} \mid \text{Time Bucket})$$

Stacked proportional bar charts were used to visualise how completion, partial, and disqualified proportions vary as a function of engagement depth. This enables the detection of behavioural regime shifts across time thresholds.

### 3.4.7 Predictive Modelling Using Time-Only Features

To evaluate whether completion time alone contains a sufficient predictive signal, a shallow decision tree classifier was trained using only the feature:

$$X = \text{Time Taken (minutes)}$$

The model was constrained to a maximum depth of 3 to preserve interpretability and prevent overfitting. A 70–30 train–test split was used. Training and test accuracy were reported to assess generalisation.

This experiment isolates the behavioural contribution of engagement duration independently of perceptual features.

## 3.5 Regression-Based Framework for Employer Attractiveness Estimation

This study frames employer attractiveness prediction as a supervised regression task, with the objective of estimating respondents' 1–10 attractiveness ratings using survey-derived demographic, educational, perceptual, and motivational features. The feature set includes institution and subject of study, education level, nationality, gender, current perception of the organisation, career motivations, and the types of employer information respondents wish to learn more about. Categorical variables were encoded using Label Encoding for compatibility with tree-based models, and a fixed set of **19 features** was consolidated into a single modelling dataset.

The data was split into 80% training and 20% testing sets using a fixed random seed. Four regression models were evaluated: Linear Regression, Ridge Regression, Random Forest Regressor, and Gradient Boosting Regressor. Model performance was assessed using **RMSE, MAE, and $R^2$** on the test set, together with **5-fold cross-validated RMSE** to evaluate generalisation robustness. Feature importance was extracted from the Random Forest model to identify the most influential predictors of employer attractiveness.

### 3.5.1 Choosing a Model to Predict the Attractiveness Level

The feature set was constructed to capture four core dimensions of respondent information: educational background, demographic attributes, organisational perception, and career-related preferences. Educational and demographic variables included institution of study, main subject of study, education level, nationality, and gender. Organisational perception was represented by respondents' current view of

the organisation as an employer. Career-related variables reflected both respondents' motivations (e.g. career growth, meaningful work, job security, compensation, and work–life balance) and their interest in learning more about specific employer attributes such as role types, career progression, compensation and benefits, work–life balance, and the application process.

Categorical variables, namely institution, subject of study, nationality, gender, and organisational perception, were encoded using Label Encoding for compatibility with tree-based models, which are insensitive to monotonic transformations of category codes and do not require one-hot expansion. All predictors were consolidated into a single modelling table, with each row representing a respondent and each column corresponding to a survey-derived feature. The final dataset consisted of a fixed feature set of 19 variables spanning institutional affiliation, academic discipline, demographic attributes, organisational perception, motivation factors, and employer information preferences.

The data was randomly split into 80% training and 20% testing sets using a fixed random seed to ensure reproducibility. Four regression models were trained and compared to capture both linear and non-linear relationships between the predictors and employer attractiveness scores: Linear Regression, Ridge Regression, Random Forest Regressor (100 trees, maximum depth = 10), and Gradient Boosting Regressor (100 estimators, maximum depth = 5).

All models were trained using the same feature set and target variable to ensure fair comparison. Performance was evaluated on the test set using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and $R^2$. To assess generalisation robustness, 5-fold cross-validated RMSE was also computed for each model.

### 3.5.2 Model Comparison and Feature Importance Analysis

All four models were assessed using RMSE (Root Mean Squared Error), MAE (Mean Absolute Error), $R^2$ Score, and 5-Fold Cross-Validation to ensure robustness.The Gradient Boosting Regressor was selected for downstream analysis as it achieved the best performance with an $R^2$ of 0.165 and RMSE of 0.213. Random Forest was a close second ($R^2$ 0.157), while linear models performed significantly worse ($R^2$ ~0.077), indicating non-linear relationships in the data.

Feature importance scores were extracted from the trained Gradient Boosting model using the impurity-based importance measure. These scores quantify each feature's relative contribution to reducing prediction error across all decision trees in the ensemble. The top 15 most influential predictors were visualised using a horizontal bar chart to support interpretability and facilitate qualitative comparison with earlier descriptive and inferential findings.

This feature importance analysis provided a complementary, model-driven perspective on which respondent attributes and perceptions most strongly influence predicted employer attractiveness, enabling direct alignment between predictive modelling outputs and earlier exploratory insights.

# 4 Results

## 4.1 Patterns Behind Survey Questions

Across the Global MCA and the combined MCA→SVD pipeline, educational identity emerged as the dominant source of differentiation. The Global MCA contribution analysis for Component 0 was strongly driven by institution and subject-area categories, with both affirmative and negative versions of institution indicators appearing as top contributors (Fig 3). In practical terms, the survey's primary dimension is

capturing an education/institution opposition structure first, and then capturing employer-attitudinal differences second. The combined feature pipeline reinforced this observation (Fig 4). The first global SVD component was most influenced by the MCA dimensions derived from the "institution attended" block, followed by education level and field-of-study dimensions, with employer attractiveness, motivations, perception, and information needs contributing at a lower but still meaningful level.

```
Top 10 contributing categories to Component  (After Filtering):0

National University of Singapore (NUS)__No                                                                    0.179693
Nanyang Technological University (NTU)__Yes                                                                   0.130078
National University of Singapore (NUS)__Yes                                                                   0.097265
Natural Science (Physics, Chemistry, Biology, Earth Sciences, etc.)__Yes                                      0.077393
Business/Management (Accounting, Finance, Marketing, Management, etc.)__Yes                                  0.051404
I'm familiar with the organisation and would consider them as a potential employer__Yes                       0.047361
Nanyang Technological University (NTU)__No                                                                    0.038731
Yale-NUS College__Yes                                                                                         0.031426
Natural Science (Physics, Chemistry, Biology, Earth Sciences, etc.)__No                                       0.028707
Engineering (Electrical/Electronics, Mechanical, Civil, Chemical, Industrial Engineering, etc.)__Yes          0.027412
Name: 0, dtype: float64
```

Figure 3

```
--- Top contributing feature dimensions (absolute loadings) ---
Which_higher_education_institution_do_you_or_did_you_study_at?_Dim0                                           0.598241
Education_level                                                                                               0.372575
Which_of_the_following_best_describes_the_main_subject_that_you_are_studying?_Dim1                            0.372142
Which_of_the_following_best_describes_the_main_subject_that_you_are_studying?_Dim0                            0.321788
Which_of_these_factors_would_most_motivate_you_to_apply_for_a_position_at_the_organisation?_Dim0              0.269202
How would you rate the attractiveness of the organisation as an employer?                                     0.248286
What_do_you_wish_to_learn_more_about_regarding_the_organisation_as_an_employer?_Dim0                          0.200180
Which_of_these_statements_best_describes_your_current_perception_of_the_organisation_as_an_employer?_Dim0    0.173240
Which_of_these_statements_best_describes_your_current_perception_of_the_organisation_as_an_employer?_Dim1    0.140365
What_is_your_gender?_Dim0                                                                                     0.137821
Name: SVD_Comp_0, dtype: float64
```

Figure 4

Local MCA provided clearer behavioural interpretation within individual questions. For information needs, option tendencies separated respondents who wanted cultural and process clarity (e.g., work-life balance/culture; application and interview process) from those focused on tangible employment details (e.g., roles available; compensation and benefits; career progression) (Fig 5). For motivators, meaningful work impact behaved as a highly discriminating signal, while work-life balance separated along a different dimension, indicating multiple motivation styles rather than a single "importance" continuum (Fig 6). Employer perception also separated cleanly, reflecting staged states from low clarity to familiarity and active consideration. Persona clustering produced three coherent segments: one with higher familiarity/consideration and meaningful impact motivation; a second dominated by NUS and Natural Sciences with weaker employer clarity and stronger interest in work-life balance, culture, and process transparency (plus higher emphasis on job security); and a third represented by NTU and SIM with business/engineering fields, characterised by learning-and-development motivation and a "positive but shallow" employer impression, suggesting conversion potential through clearer role and pathway information (Fig 7).

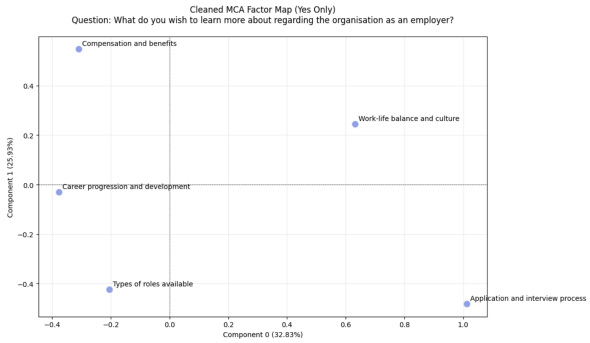


Figure 5

Figure 6

Figure 7

Finally, the semantic redundancy analysis confirmed high-confidence overlaps. “Career progression and development” was highly similar to “Career growth opportunities,” and “Work-life balance and culture” was very close to “Work-life balance,” indicating duplicated measurement of the same constructs. These results directly inform targeted survey improvements.

## 4.2 Geographical cluster profiling

Geographical clusters are compared in terms of motivational patterns, learning preferences, and employer perceptions.

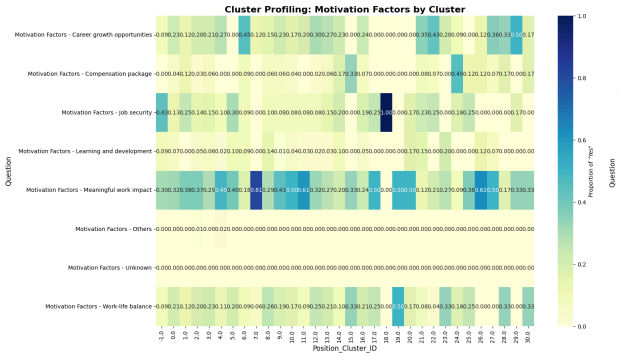


Figure 8

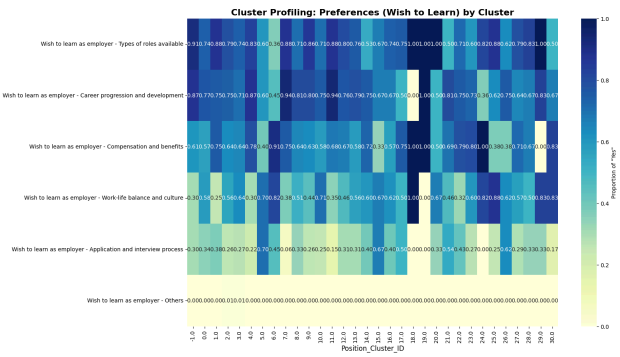


Figure 9

**Motivational Differentiation (Fig 8):** Career growth opportunities concentrate in clusters 5, 10, and 23, while meaningful work impact peaks in cluster 7. Job security dominates cluster 18, distinguishing it sharply from other segments. Work-life balance shows more even distribution, with slight elevation in clusters 19 and 27. Compensation remains relatively uniform, suggesting it functions as a baseline rather than differentiating factor.

**Learning Preferences (Fig 9):** "Types of roles available command" universal interest (>70%), peaking in clusters 11, 18-20, and 30. "Career progression" shows similarly high engagement (60-90%), strongest in clusters 11, 18-19, and 30. "Work-life balance" information varies substantially. Clusters 1, 4-5, and 11 show lower interest compared to clusters 18-19. Application process information follows an inverse pattern, with clusters 1-4 and 27-30 showing weaker engagement.

**Attractiveness and Familiarity:** Perceived attractiveness varies considerably by cluster. Cluster 23 exhibits exceptionally high ratings (>0.9), while clusters 7, 10, 15, 18, and 27 rate moderately high (0.7-0.78). Clusters 4, 12, 20-21 show markedly lower scores (0.4-0.5), indicating potential brand misalignment. Employer familiarity follows similar patterns; clusters 23-24 show 80-90% positive consideration, while clusters 1-8 and 11-17 predominantly have vague positive impressions without detailed knowledge.

**Demographics:** Gender composition varies from near-parity (clusters 4, 5, 20) to heavily male-dominated (clusters 7, 23-24 at >85%). Most clusters are Singaporean/PR-majority (>85%), except clusters 13, 18, and 25 with 20-65% non-Singaporean representation. Field of study drives clear differentiation: Business concentrates in clusters 0-1, 4, 9-12, 18; Engineering dominates clusters 7, 22, 27; Natural Sciences appear strongly in clusters 1, 3, 7, 9, 11-12, 19-20.

## 4.3 Employer Attractiveness Drivers

The Gradient Boosting model identified current organizational perception as the strongest predictor of employer attractiveness (importance: 0.23), followed by field of study (0.18) and educational institution (0.16). Education level and gender showed moderate predictive power (0.09 and 0.05), while learning preferences (types of roles, career progression, work-life balance) each contributed 0.03-0.04. Motivational factors demonstrated modest individual importance (0.02-0.03 each), though their collective contribution remains meaningful. (Fig 9)

Attractiveness scores varied substantially across disciplines. Historical/Philosophical Science, Politics, and Law students rated highest (0.70-0.75), followed by Economics, Psychology, and Architecture (0.63-0.65). Engineering, IT, and Mathematical Science students clustered near the overall mean (0.59-0.61). Business/Management and Media students rated slightly below average (0.56-0.58), while

Natural Sciences, Medicine, Creative Arts, and Geography students exhibited the lowest scores (0.48-0.54), suggesting potential misalignment with these disciplines' priorities. (Fig 10)

James Cook University, Curtin University, and SMU students rated the organization highest (0.67-0.71), well above the 0.59 mean. NTU, SIT, and SIM students showed moderately elevated scores (0.62-0.65). NUS students rated slightly below average (0.54), while NIE, NAFA, and LASALLE students demonstrated lower perceptions (0.47-0.49). Yale-NUS students showed the lowest scores (0.32). (Fig 11)

Strong correlation exists between familiarity and attractiveness. Respondents familiar with the organization who would consider it as an employer rated attractiveness highest (0.70, n=374), while those with general positive impressions rated 0.60 (n=555). Name recognition without clear impressions yielded 0.50 (n=564), and unfamiliarity resulted in 0.30 (n=355). This 0.40-point spread underscores the critical role of employer brand awareness in shaping attractiveness perceptions. (Fig 11)

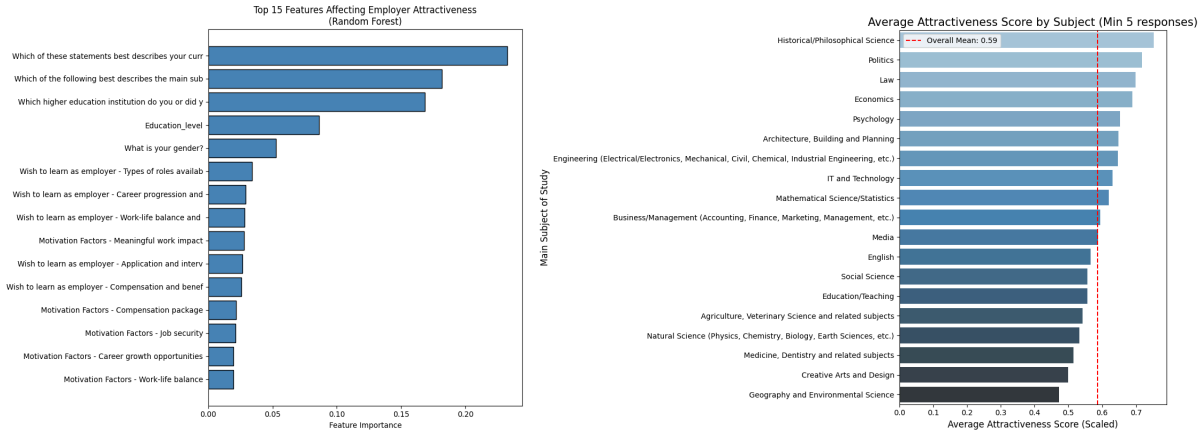


Figure 10

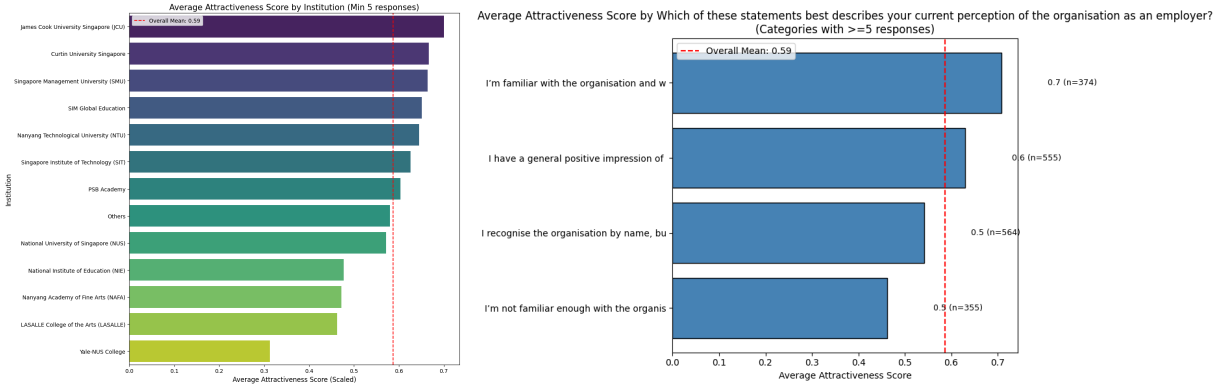


Figure 11

## 4.4 Partial Response and Dropout Behaviour

The logistic regression results show that informational intent is the strongest driver of survey completion. Variables capturing what respondents wished to learn about the organisation, such as types of roles, career progression, compensation and benefits, work–life balance, and the application process, exhibited the largest odds ratios, indicating several-fold increases in completion likelihood. In contrast, general

employer attractiveness ratings and most intrinsic motivation factors had much weaker or negligible effects.

Time-based analysis reveals a strong behavioural separation across survey outcomes. Completed responses had a higher median completion time (1.52 minutes) than disqualified (0.77 minutes) and partial (0.65 minutes) responses, indicating that sustained engagement is rare but highly diagnostic. A Kruskal–Wallis test confirmed that completion-time distributions differ significantly across groups ($H = 415.868$, $p < 0.00001$, $\eta^2 \approx 0.173$), with all pairwise differences remaining significant after correction.

Time-bucket analysis identified a clear behavioural break at the 1-minute mark. Fewer than half of respondents completed the survey if they disengaged within the first minute, while completion rates exceeded 85–95% beyond this threshold, and partial responses became negligible after 3 minutes. A shallow decision tree trained using only completion time achieved 79.7% test accuracy, showing that simple time-on-task thresholds are sufficient to predict most survey outcomes.

### 4.5 Discussion

The results show that the proposed framework effectively addresses the key objectives of the SDS Datathon. Respondent segmentation reveals meaningful demographic and behavioural heterogeneity, supporting persona-driven employer branding. Driver analysis uncovers interpretable determinants of employer attractiveness, while partial-response diagnostics highlight concrete opportunities to reduce survey friction and improve completion rates.

Although clustering separation is moderate due to overlapping human preferences, the resulting segments remain practical and interpretable. Overall, the findings demonstrate the feasibility and real-world value of a data-driven internal analytics tool for survey optimisation and employer-attractiveness intelligence.

# 5 Insights

This section summarises the key insights derived from the proposed survey analytics framework. These insights focus on analysis of respondent profile, how survey design can be improved, and how the extracted patterns can support internal decision-making for surveyors and employer-branding teams.

## 5.1 Insights from Data-Driven Analysis

### 5.1.1. Respondent Profile Is Dominated by Perception and Educational Identity

Global latent-structure analysis shows that respondents' current perception of the organisation and their educational identity (institution and field of study) are the strongest drivers of employer attractiveness. In Global MCA, Component 0 is driven primarily by institution and subject categories together with employer familiarity ("familiar and would consider"), while variables such as nationality contribute essentially zero explanatory power (Fig 12). This indicates that baseline brand perception and academic alignment shape attractiveness more strongly than demographic attributes or isolated motivation factors. More importantly, education identity and broad perception which carry most of the explanatory power should be retained, while nationality which is currently mis-specified (too sparse or uninformative) should be redesigned using higher-level groupings or routing to avoid collapsing to zero contribution.

```
Top 10 contributing categories to Component  (After Filtering):0

National University of Singapore (NUS)__No                                                    0.179693
Nanyang Technological University (NTU)__Yes                                                   0.130078
National University of Singapore (NUS)__Yes                                                   0.097265
Natural Science (Physics, Chemistry, Biology, Earth Sciences, etc.)__Yes                      0.077393
Business/Management (Accounting, Finance, Marketing, Management, etc.)__Yes                   0.051404
I'm familiar with the organisation and would consider them as a potential employer__Yes       0.047361
Nanyang Technological University (NTU)__No                                                    0.038731
Yale-NUS College__Yes                                                                         0.031426
Natural Science (Physics, Chemistry, Biology, Earth Sciences, etc.)__No                       0.028707
Engineering (Electrical/Electronics, Mechanical, Civil, Chemical, Industrial Engineering, etc.)__Yes   0.027412
Name: 0, dtype: float64


Least contributing categories to Component  (After Filtering):0

Singaporean/Singapore PR__Yes                                1.995841e-63
Application and interview process__No                        1.220498e-05
Prefer not to say__No                                        1.365956e-05
Agriculture, Veterinary Science and related subjects__No     2.457390e-05
Application and interview process__Yes                       3.003247e-05
Architecture, Building and Planning__No                      3.077528e-05
Geography and Environmental Science__No                      3.172570e-05
Creative Arts and Design__No                                 3.418670e-05
Education/Teaching__No                                       3.658238e-05
Career growth opportunities__No                              4.307853e-05
Name: 0, dtype: float64
```

Figure 12

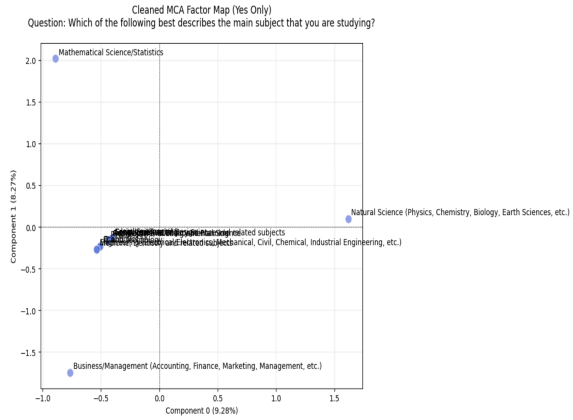


Figure 13

### 5.1.2. Strong Evidence of Semantic and Behavioural Redundancy Across Survey

Both behavioural co-occurrence patterns and embedding-based semantic similarity reveal near-duplicate constructs across multiple question blocks. For example, "Work-life balance and culture" overlaps strongly with "Work-life balance" (≈0.83), and "Career progression and development" overlaps with "Career growth opportunities" (≈0.77). These high similarity scores indicate that the same respondent priorities are being measured repeatedly under different labels, violating a "one construct, one label" principle and introducing redundancy and ambiguity into survey interpretation.

### 5.1.3. Insights from Geographical Cluster Analysis

The geographical cluster analysis reveals three strategically distinct segments requiring differentiated recruitment approaches. High-engagement clusters (18-20, 23, 30) demonstrate exceptional employer brand resonance with attractiveness exceeding 0.9 and 80-90% positive consideration, indicating recruitment-readiness where conversion-focused strategies would yield optimal returns. Awareness-gap clusters (1-8, 11-17) present untapped potential with moderate attractiveness (0.5-0.6) but limited organizational knowledge, suggesting high ROI from targeted brand-building investments given their universal interest in role types (>70%). Misaligned clusters (4, 12, 20-21) warrant strategic reassessment, as low attractiveness scores (0.4-0.5) despite adequate familiarity indicate fundamental value proposition disconnection. Motivational hierarchies differ markedly, for example cluster 18 prioritizes job security, clusters 5/10/23 emphasize career growth, and cluster 7 values meaningful impact, while compensation functions uniformly as a baseline expectation rather than differentiator. Information consumption patterns split between comprehensive researchers (clusters 11, 18-20, 30 with 60-90% engagement) and selective seekers (clusters 1-4, 27-30) who disengage from application details, suggesting need for barrier-reduction messaging. Demographic patterns reveal gender imbalances correlating with attractiveness (male-dominated clusters 7, 23-24 at >85% show high ratings), substantial international student concentration in clusters 13, 18, 25 (20-65% non-Singaporean), and clear discipline-based clustering with Business concentrating in clusters 0-1, 4, 9-12, 18 and Engineering dominating clusters 7, 22, 27.

## 5.2 Practical Implications for Survey Design and Employer Branding

### 5.2.1. Enforce a "One Construct, One Label" Design Principle

Near-synonymous constructs such as career growth vs. career progression and work–life balance vs. work–life balance and culture should be merged into single canonical options. Each conceptual dimension should appear only once in the survey, either as a motivator or as an information need, but not under multiple labels. This reduces cognitive load, eliminates ambiguity, and preserves measurement validity while shortening the survey. Strategies to achieve "One Construct, One Label" include the following:

### 5.2.2. Preserve High-Lift Segmentation Anchors and Remove Low-Value Blocks

Each persona is defined by multiple high-lift signals across different blocks, confirming that segmentation is not driven by a single question (Fig 7). For example, Persona 1 is anchored by Natural Science (2.56) and NUS (1.50), and is also marked by low employer clarity ("recognise name but unclear impression," 1.60) and stronger need for reassurance/clarity (learn-more: "work-life balance and culture," 1.15; "application and interview process," 1.08). The implication is to preserve high-lift items as core segmentation anchors, and when an anchor is broad (e.g., "application process"), add a targeted follow-up that identifies the specific gap rather than introducing more broad, overlapping questions.

### 5.2.3. Option Reduction and Refinement by Local MCA Geometry

Options can be reduced by keeping categories that sit at extremes (high separation) and merging those that cluster closely (redundant behaviour). In the main subject question, Natural Science (Dim0 0.991), Mathematical Science/Statistics (Dim1 0.834), and Business/Management (Dim1 -0.675) should remain standalone because they drive separation, while Engineering, Medicine, IT/Technology, and Economics cluster near each other and can be merged into an "Applied STEM" group (Fig 13). Similarly, Psychology, Geography/Environmental Science, etc., cluster tightly and can be merged into "Other disciplines." This yields a shorter subject list such as Natural Science / Math-Stats / Business-Management / Applied STEM / Other disciplines with minimal loss of segmentation power. The same geometry-based reduction principle can be applied across other multi-option blocks to remove redundant categories while preserving the options that genuinely differentiate respondents.

### 5.2.4. Strategic Segmentation for Targeted Recruitment

Effective recruitment cannot rely on a one-size-fits-all approach. Instead, organizations must tailor their strategies to different geographical segments by emphasizing the specific motivations each values most, whether job security, career growth opportunities, or meaningful work impact. Resources should be strategically allocated to high-potential clusters based on their current levels of employer attractiveness and brand familiarity, rather than spreading efforts uniformly across all markets.

# 6 Conclusion & Future Work

This study presents a multi-stage survey analytics framework that transforms raw GRADSG survey responses into actionable and interpretable insights for survey optimisation and employer-attractiveness intelligence. By integrating co-occurrence analysis, semantic similarity scoring, respondent segmentation, and latent-structure modelling, the framework operationalises the core objectives of the SDS Datathon: detecting correlated and redundant questions, uncovering meaningful respondent personas, and extracting applicant-centric drivers of employer attractiveness.

The results show that employer attractiveness is dominated by brand perception and educational identity, while several demographic and process-related variables contribute negligible explanatory value.

Embedding-based similarity and behavioural co-occurrence patterns provide strong evidence of near-duplicate survey constructs, while local MCA geometry reveals that only a subset of answer options genuinely differentiates respondents. Respondent personas are shown to be multi-signal and structurally coherent, supporting persona-driven employer branding and targeted communication strategies.

Several limitations remain. Clustering separation is moderate due to overlapping human preferences, free-text responses are analysed using relatively lightweight methods, and most findings remain correlational rather than causal.

Future work should focus on improving segmentation through representation learning, incorporating causal inference to isolate the effects of employer attributes, and applying deeper NLP methods to free-text responses. From a product perspective, the framework could be deployed as an interactive internal dashboard with automated survey-design recommendations such as option mergers, block removals, and question reordering. Together, these extensions would transform the prototype into a scalable internal intelligence tool for evidence-based survey optimisation and employer-branding strategy.